

Predictive Delinquency Risk Model – Business Summary Report

Prepared for: Head of Collections, Geldium | Data source: 500-customer account dataset (Task 1 EDA)

1. Summary of Predictive Insights

Across the 500-account dataset, delinquency (10–12% of accounts) is driven far more by behavior than by demographics or income. Three variables stand out:

- **Missed payments** – the strongest single predictor (correlation +0.48). Customers with 3 or more missed payments in the last 6 months are the clearest early-warning group.
- **Credit utilization** – the second strongest driver (correlation +0.33). Risk rises sharply once utilization passes 70% of available credit.
- **Credit score** – a meaningful negative predictor (correlation –0.30); scores below ~450, or a drop of 50+ points, warrant review.

Debt-to-income ratio and account tenure add secondary signal, while income has only a weak, indirect relationship with delinquency once behavior is accounted for – income should not be used as a stand-alone screen.

Key Insight	Customer Segment	Influencing Variables	Potential Impact
Repeated missed payments most reliably predict delinquency	Accounts with ≥ 3 missed payments in 6 months	Missed_Payments, Month_1–Month_6 status	Early outreach before the account becomes delinquent
High utilization signals customers living beyond means	Accounts with utilization > 70%	Credit_Utilization, Loan_Balance	Proactive credit-limit or hardship conversation
Low / falling credit score signals weakening creditworthiness	Score < 450 or a 50+ point drop	Credit_Score	Trigger manual account review

Top 3 Risk Factors: (1) missed payments, (2) high credit utilization, (3) low/declining credit score.

2. Recommendation Framework

Restated Insight: Customers with 3+ missed payments in the last 6 months and credit utilization above 70% are the clearest early-warning combination for delinquency risk.

Proposed Recommendation: Launch a proactive outreach and support program for customers the model flags as High Risk, offering payment reminders and a fast path to a short-term hardship or debt-management plan – before the account becomes delinquent.

Specific

Automated SMS/email reminders plus a live-agent offer of a short-term hardship plan, targeted at customers flagged High Risk (Missed_Payments ≥ 3 OR Utilization >70% OR Credit_Score <450).

Measurable

Track 30/60/90-day delinquency rate and hardship-plan enrollment for the pilot group versus a matched control group; target a 10–15% relative reduction in new delinquencies among flagged customers.

Actionable

Uses fields already in the data (Missed_Payments, Credit_Utilization, Credit_Score) and existing Collections contact channels – only threshold rules from the model need to be applied to the outreach queue.

Relevant

Targets the three strongest, model-validated predictors of delinquency and directly supports Collections' goal of reducing delinquency while preserving the customer relationship.

Time-bound

6-week pilot, with a go/no-go review at week 6 for portfolio-wide rollout.

Justification and Business Rationale

This action is scalable (rules-based flagging extends to the full portfolio once validated), easy to implement (uses existing Collections workflows), fair (reviewed via the fairness audit across Location, Employment_Status, Age and Card_Type so outreach isn't skewed toward any one group), and cost-effective (early, supportive outreach is materially cheaper than post-delinquency collections). It is directly grounded in the data: missed payments and utilization are the two strongest correlated predictors of delinquency identified in the analysis.

3. Ethical and Responsible AI Considerations

Fairness risks and mitigations

- **Missing-data bias** – Income (7.8% missing) and Loan_Balance (5.8% missing) could be imputed inaccurately for specific segments (e.g., self-employed customers). Mitigation: use group-based/KNN imputation and check whether missingness itself correlates with a segment before trusting imputed values.
- **Proxy-variable bias** – Location (5 cities) or Credit_Card_Type could stand in for income or socioeconomic status. Mitigation: run the model's Fairness Audit step to compare error rates across these groups, and use SHAP to confirm predictions rely on behavior (missed payments, utilization) rather than these proxies.

Explainability

The champion model is paired with SHAP so the Collections team can see which factors (e.g., missed payments, utilization) drove any individual customer's risk score, in plain language – supporting clear, defensible conversations with customers.

Responsible decision-making

The recommended action is supportive (outreach and hardship options) rather than punitive (e.g., automatic credit-limit cuts), and Moderate/High-risk cases are routed to human review rather than triggering automated account actions. Customer financial data is protected throughout, and the model should be re-audited periodically as customer behavior and economic conditions change.